Automated Health Claim Fact-Checking

RAG vs Multi-Agent Approaches

CS614 Generative AI | Group 2 | March 2026

SciFact Benchmark

5,183
Abstracts

300
Test Claims

9+
Experiments

82%
Best Accuracy

The Problem: Health Misinformation at Scale

Infodemic

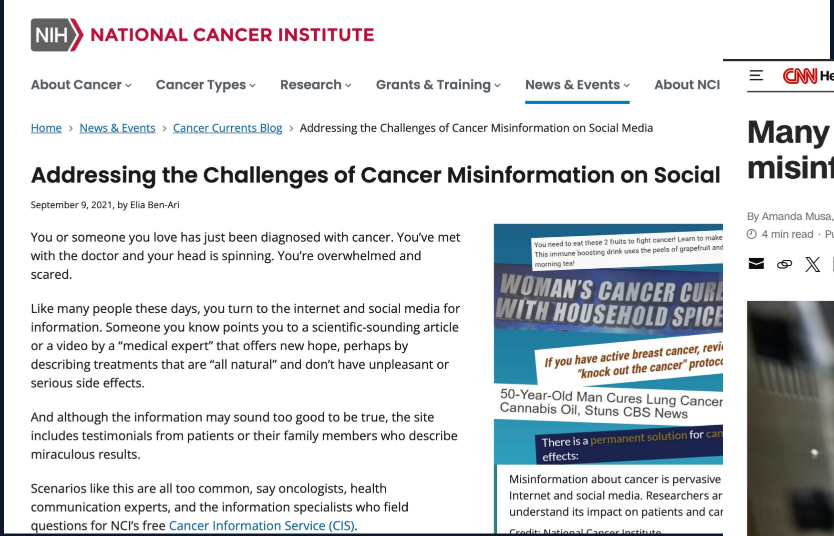
WHO declared alongside COVID-19

15–30 min

per claim — manual fact-checking

Overwhelmed

fact-checkers on Meta & X


NIH NATIONAL CANCER INSTITUTE

About Cancer Cancer Types Research Grants & Training News & Events About NCI

Home › News & Events › Cancer Currents Blog › Addressing the Challenges of Cancer Misinformation on Social Media

Addressing the Challenges of Cancer Misinformation on Social

September 9, 2021, by Elia Ben-Ari

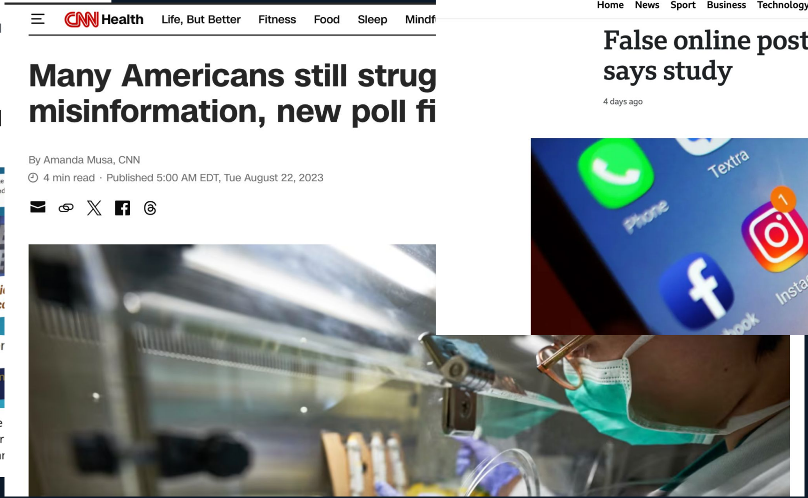
You or someone you love has just been diagnosed with cancer. You've met with the doctor and your head is spinning. You're overwhelmed and scared.

Like many people these days, you turn to the internet and social media for information. Someone you know points you to a scientific-sounding article or a video by a "medical expert" that offers new hope, perhaps by describing treatments that are "all natural" and don't have unpleasant or serious side effects.

And although the information may sound too good to be true, the site includes testimonials from patients or their family members who describe miraculous results.

Scenarios like this are all too common, say oncologists, health communication experts, and the information specialists who field questions for NCI's free Cancer Information Service (CIS).

Misinformation about cancer is pervasive Internet and social media. Researchers ar understand its impact on patients and car


CNN Health Life, But Better Fitness Food Sleep Mindf

Many Americans still strug misinformation, new poll fi

By Amanda Musa, CNN

4 min read · Published 5:00 AM EDT, Tue August 22, 2023


BBC

Home News Sport Business Technology Health Culture Arts Travel Earth Audio Video Live

False online posts fuel self-diagnosis, says study

4 days ago

Share Save

The Problem: Health Misinformation at Scale

Infodemic

WHO declared alongside COVID-19

15–30 min

per claim — manual fact-checking

Overwhelmed

fact-checkers on Meta & X

Our Goal

Build and compare automated pipelines that verify health claims against scientific evidence. Evaluate the tradeoff between simple RAG and complex agentic architectures — not just which is more accurate, but when each approach is the right tool.

"Vitamin D prevents COVID-19"

"Vaccines cause autism"

"Aspirin prevents heart attacks"

↑ Claims like these spread faster than human fact-checkers can respond

Dataset: SciFact Benchmark (Allen AI, EMNLP 2020)

Corpus: 5,183 Scientific Abstracts

Avg abstract length: **1,404 chars** | Biomedical: genetics, immunology, oncology, epidemiology
19% have structured sections (Background / Methods / Results / Conclusion)

Test Claims: 300 (Perfectly Balanced)

100	100	100
SUPPORTED	UNSUPPORTED	INSUFFICIENT

Easy vs Hard Split — Critical for Later Analysis

200 EASY Claims

Gold evidence document EXISTS in corpus
System should find it with good retrieval

100 HARD Claims

NO gold evidence in corpus
System must reason from tangential papers

The hard/easy split explains why agents underperform on SciFact but may excel in real-world open-domain scenarios

System Architecture: Three Experimental Axes

Health Claim — Input → **AXIS 1: CHUNKING** — How we split 5,183 abstracts → **ChromaDB Index** — all-MiniLM-L6-v2 (384-dim) → **AXIS 2: RETRIEVAL** — How we find relevant evidence → **AXIS 3: REASONING** — LLM interprets evidence → **Verdict** — SUP / UNS / INS

Tech Stack

Embeddings	LLM	Agents	Storage	External
Sentence-Transformers all-MiniLM-L6-v2	Claude Sonnet 4 Anthropic API + AWS Bedrock	AWS Strands SDK LangGraph (LangChain)	ChromaDB Persistent vector store	Semantic Scholar API 200M+ papers

Axis 1: Chunking Strategies

Fixed

Splits abstract into ~150 words with ~38-word overlap.

Section-Aware

Uses regex to split sections (e.g. Aim, Method) first before fixed chunking. Falls back to fixed chunking if unstructured.

Semantic

Splits when cosine similarity between sentences < 0.5 first before fixed chunking.

Recursive

Tries paragraph → sentence → space → characters. Targets ~200-word windows with ~38-word overlap.

Metadata Injection: doc_id, title, chunk_index, structured, (section)

Chunking Example

doc_id: 1986482

title: The Impact of the New WHO Antiretroviral Treatment Guidelines on HIV Epidemic Dynamics and Cost in South Africa

abstract: BACKGROUND Since November 2009, WHO recommends that adults infected with HIV should initiate antiretroviral therapy (ART) at CD4+ cell counts of …

structured: true

Axis 1: Chunking Example

Fixed (79.9%)

[0] BACKGROUND Since November 2009, WHO recommends that adults infected…
[1] ***≤200 cells/μl for the next 30 years.*** During the first five years…
[2] ***ART prices and effectiveness.*** Therefore, South Africa should aim…

✓ Fast, consistent, easy to implement
✗ Semantically blind, leads to incoherent chunks

Section-Aware (77.3%)

[Aim] Since November 2009, WHO recommends that adults infected…
[Method] ***AND FINDING*** We used an established model of the transmission…
[Method] ***average 16 years.***
[Conclusion] Our study strengthens the WHO recommendation of starting ART…

✓ Preserves document structure
✗ 81% unstructured, can miss or fragment sections

Semantic (76.3%)

[0] BACKGROUND Since November 2009, WHO recommends that adults infected…
[2] ***We quantified the model to represent Hlabisa subdistrict, KwaZulu-Natal, South Africa.***
[7] Therefore, South Africa should aim…

✓ Semantically meaningful chunks
✗ Slow, can fragment short abstracts

Recursive (82%)

[0] BACKGROUND Since November 2009, WHO recommends that adults infected…
[1] METHODS AND FINDING We used an established model of the transmission…
[2] CONCLUSIONS Our study strengthens the WHO recommendation of starting ART…

✓ Well-formed chunks without the need of embeddings

Recursive wins Used for all subsequent experiments

Axis 2: Retrieval Methods

01

Naive (Dense Embedding)

- Query claim directly against ChromaDB
- Cosine similarity search
- Return top-5 passages
- ~100ms per query

02

Hybrid (Dense + BM25)

- Combine embedding sim with BM25 keyword matching
- BM25 catches exact medical term matches
- Reciprocal Rank Fusion to merge ranked lists
- Return top-10, truncate to top-5

03

Hybrid + Cross-Encoder Rerank

- Same as hybrid, but add cross-encoder pass
- Cross-encoder scores (query, passage) pairs jointly
- More accurate but slower than bi-encoder
- Return top-5 after reranking

Hypothesis: BM25 would catch exact gene/drug names that embedding similarity might miss. Reranking would fix hybrid's ranking quality. In practice — neither helped on this biomedical corpus.

Axis 2 Results: Retrieval Comparison

All experiments use Recursive chunking (best from Axis 1)

Method	Accuracy	Avg Latency	Cost / 300 Claims
Naive (Dense Embedding) ★ WINNER	**82.0%**	8.4s	$3.46
Hybrid (Dense + BM25)	79.0%	6.1s	$3.60
Hybrid + Cross-Encoder Rerank	81.3%	6.0s	$3.72

Why did Naive win?

BM25 False Positives

Keyword matches on common medical terms ("patients", "treatment") that weren't semantically relevant to the specific claim

Reranking Recovered 2.3pp

79.0% → 81.3% by filtering BM25 noise — but didn't surpass pure embedding search

Focused Corpus Effect

SciFact is all biomedical. Dense embeddings capture semantic similarity well in this domain — no need for keyword fallback

Insight: Hybrid retrieval is designed for broad, heterogeneous corpora. On a focused domain corpus, simpler is better.

Axis 3: Agent Architectures — Single-Pass vs Multi-Agent

Single-Pass RAG (E1–E6)

1 LLM call per claim

- Claim input
- Retrieve top-5 passages from ChromaDB (cosine similarity)
- Single LLM call sees all evidence at once
- Verdict: SUP / UNS / INS

Multi-Agent Pipeline (E7/E8)

4 LLM calls per claim (minimum)

01 Claim Parser	Decomposes into 2-4 sub-claims + PubMed queries
02 Retrieval Agent	Searches ChromaDB per sub-claim query
03 Evidence Reviewer	Flags contradictions, gaps, evidence quality
04 Verdict Agent	Synthesizes review → final verdict + citations

Strands vs LangGraph: Framework Comparison

E7: Strands SDK (AWS)

- LLM has tool-calling agency over retrieval — can retry and adapt searches
- AWS Bedrock as LLM backend with built-in retry + exponential backoff
- Pydantic models for structured output (RetrievalOutput, ReviewedEvidence)
- 0 connection errors in full experiment run

E8: LangGraph (LangChain)

- Deterministic StateGraph DAG — fixed retrieval per sub-claim, no adaptation
- Retrieval node calls ChromaDB directly (no LLM tool-calling for retrieval)
- State flows through typed PipelineState dictionary
- 51 BrokenPipeErrors fixed with retry wrapper — NOT a LangGraph limitation

Strands vs LangGraph: How the Pipelines Actually Work

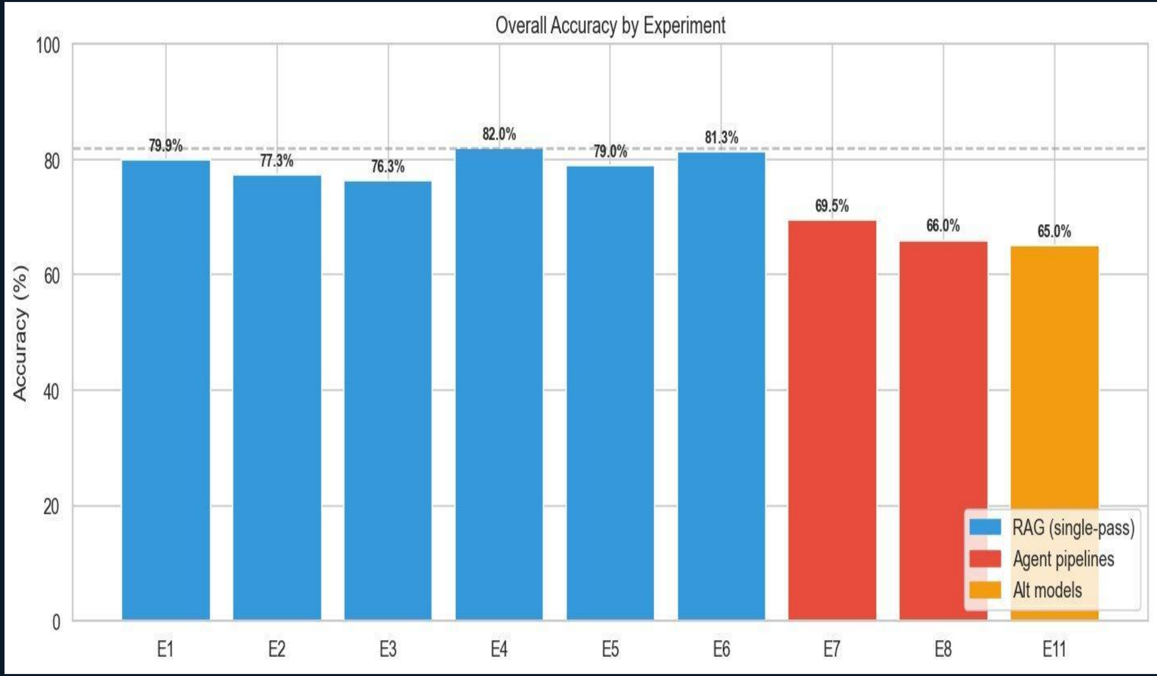
KEY DIFFERENCE

VS

E7: Strands SDK (AWS) 69.5%	E8: LangGraph (LangChain) 66.0%
Claim Input	**Claim Input**
Claim Parser Agent Decomposes → 2-4 sub-claims + PubMed search queries	**Node 1: Claim Parser** Same decomposition as E7
Retrieval Agent ← LLM AGENCY @tool search_local_corpus() — LLM decides when + how many times to call	**Node 2: Retrieval ← DETERMINISTIC** 1 ChromaDB search per sub-claim. No LLM agency. No retry.
Evidence Reviewer Agent Flags contradictions, gaps, evidence quality (RCT vs observational)	*fixed edge* **Node 3: Evidence Reviewer** Reads PipelineState → returns updated state dict
Verdict Agent SUPPORTED / UNSUPPORTED / INSUFFICIENT_EVIDENCE	*fixed edge* **Node 4: Verdict** SUPPORTED / UNSUPPORTED / INSUFFICIENT_EVIDENCE

KEY FINDING: E7 (69.5%) ≈ E8 (66.0%). LLM agency over retrieval didn't fix **query dilution**. Both underperform RAG for the same upstream reason.

Axis 3 Results: RAG vs Agents — Full Results (300 Claims)



Model quality gap: Claude Sonnet RAG (82%) vs Llama RAG (65%) = 17pp - bigger than RAG vs Agent gap (12pp)

82.0% — **E4 (RAG)**
Best overall

69.5% — **E7 (Strands)**
~80s latency

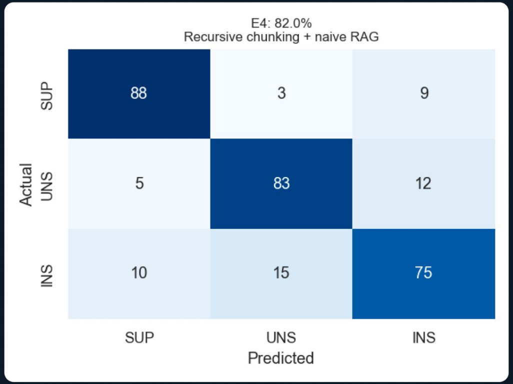
66.0% — **E8 (LangGraph)**
~100s latency

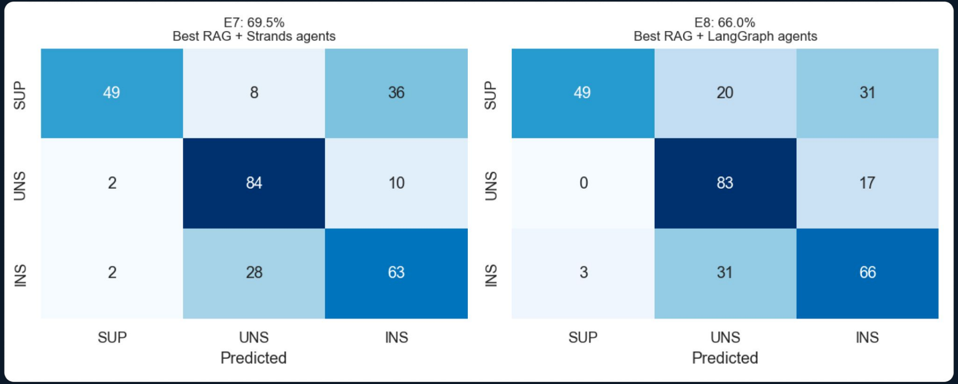
65.0% — **E11 (Llama 3.1 8B)**
~12s, ~$0 local

SUPPORTED recall: RAG 88% → Agents 49%

Reading the Results: What the Confusion Matrices Show

Before explaining why agents underperform — here's what the data actually looks like.





E4 (RAG) — Strong diagonal

88 / 83 / 75 on the diagonal. Balanced across all 3 classes : it gets most things right.

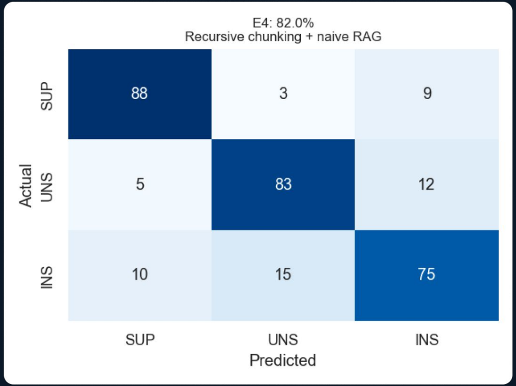
E7 / E8 — SUPPORTED row breaks

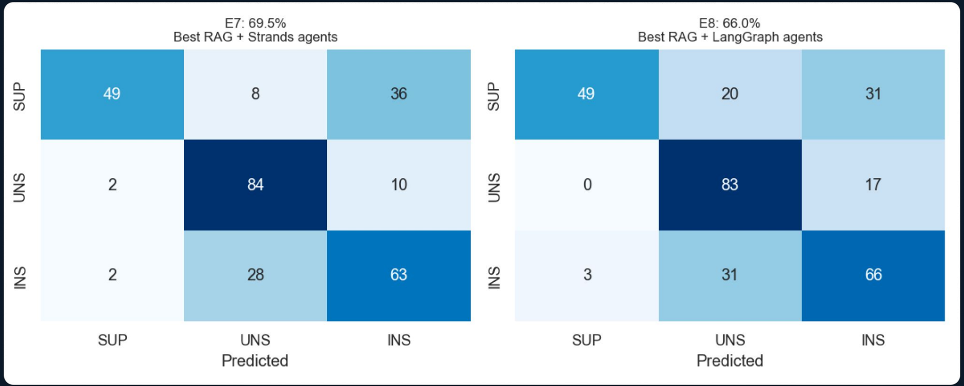
E7: 49/100 correct SUP ⟶ 36 misclassified as INS
E8: 49/100 correct SUP ⟶ 31 misclassified as INS
Both agents share the same failure pattern — when evidence supports a claim, agents miss it and default to INSUFFICIENT.

UNS detection is preserved

E7: 84, E8: 83 on UNSUPPORTED — agents catch contradictions nearly as well as RAG (83).

Root Cause Analysis: Why Agents Underperform





01 Query Dilution

Decomposed sub-claim queries are more generic — worse retrieval on well-scoped SciFact claims

02 Error Accumulation

4 sequential LLM calls × ~5-10% error rate ⟶ 0.95^4 = 81% — up to 19% accuracy loss from chaining

03 SUPPORTED Recall Collapse

88% ⟶ 49%: Diluted queries ⟶ weaker evidence ⟶ reviewer flags gaps ⟶ cautious verdict. UNSUPPORTED preserved (83-84%)

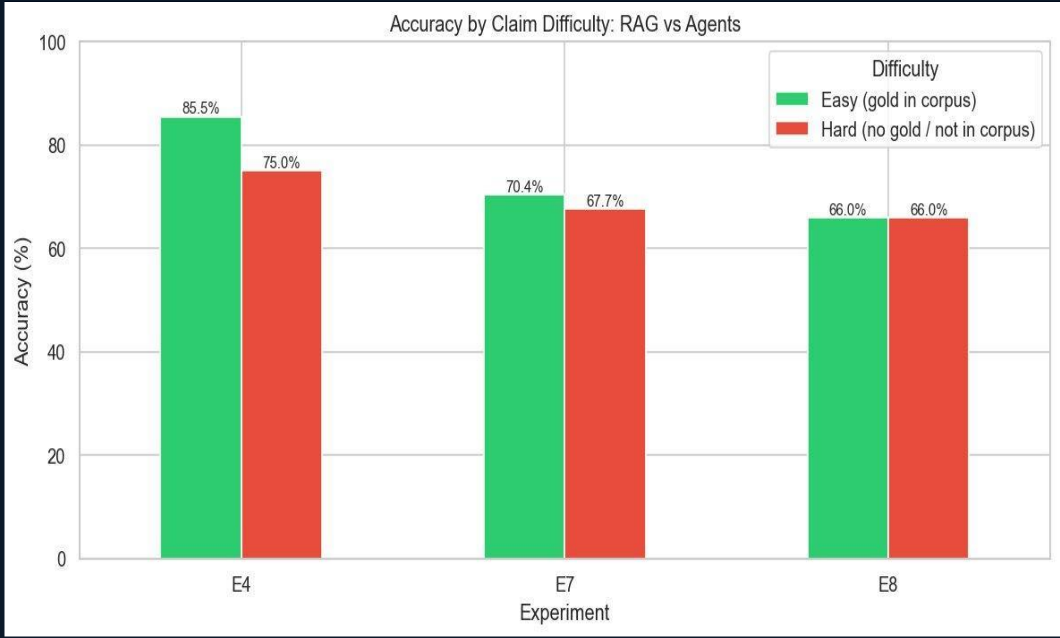
04 E7 ≈ E8 After Bug Fixes

Framework differences are minor. Bottleneck is upstream query dilution, not orchestration.

The insight is NOT 'agents are bad' — agents add value when they can take actions simple RAG cannot. Next slides show where they shine.

RAG Wins — But Has a 10.5% Blind Spot

RAG dominates on overall accuracy, but its performance is tied to one critical variable: whether the evidence is in the corpus.



Why RAG drops : 10.5% on hard claims

Hard claims = no gold evidence in corpus.
RAG's retrieval finds only tangential papers. It has no basis to confirm or contradict — so it guesses. Without the right document, accuracy collapses.

Why agents barely drop (−2.7% / 0pp)

Agents already decompose and search broadly.
They're accustomed to imperfect retrieval
the multi-step pipeline is designed to work around weak evidence. Hard claims are closer to their natural operating mode.

Implication: The 10.5% easy→hard gap is where agents can add value. On closed, well-indexed corpora RAG is unbeatable. But when the evidence runs thin — or isn't there at all — agents can bridge the gap.

What Happens When Topics Aren't in the Corpus At All?

SciFact Hard Claims vs Truly Out-of-Corpus

EASY
RAG: 85.5%

HARD
RAG: 75.0%

OUT-OF-CORPUS
RAG: 22.2%

Corpus coverage degrades ⟶

89% of out-of-corpus claims predicted INSUFFICIENT by RAG — it has no basis to reason beyond what's indexed

This is the regime where agents can justify their cost

RAG's failure mode

When no relevant evidence exists, RAG defaults to INSUFFICIENT — it can't go looking elsewhere. It's bounded by the index it was given.

Agents can go external

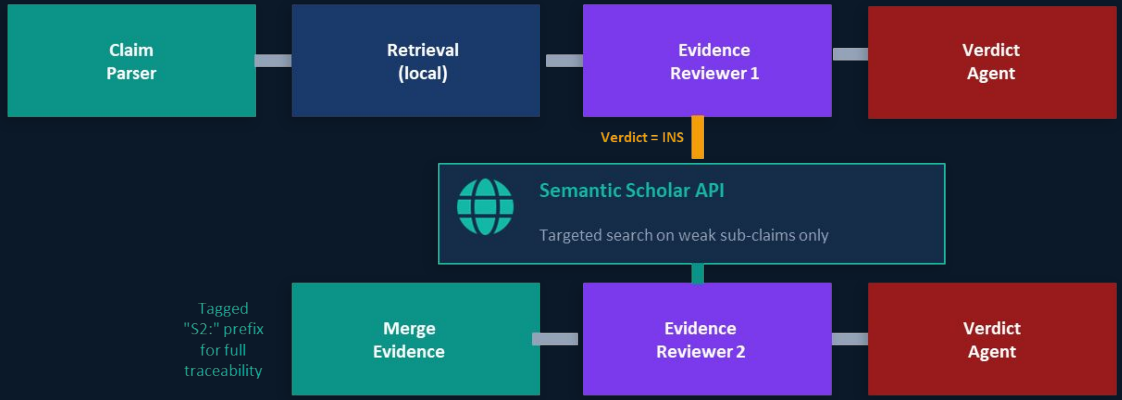
An agent with access to Semantic Scholar's 200M+ papers isn't bounded by the local corpus. It can query for evidence that was never indexed locally.

⟶ **How does an agent (E9c) compare?**

Multi Agent Architectures

Sequential Pipeline (E7/E8)

Semantic Scholar Fallback (E9c)



Sequential Pipeline

Every agent runs exactly once with no ability to recover from weak evidence.

Used as the baseline to isolate the effect of chunking strategy, retrieval method, and framework choice.

Semantic Scholar Fallback

When local evidence is weak, the system reaches out to Semantic Scholar for external papers, then re-reviews before issuing a verdict.

Solves the root problem E7/E8 couldn't as the corpus didn't have the papers.

What the Pipeline Actually Outputs — Three Real Claims

Same claim. Different architectures. Different failure modes.

Claim & Expected	E4 (RAG)	E9c (Agent + S2)
"APOE4 is not a risk factor for Alzheimer's disease." Expected: **UNSUPPORTED**	**INSUFFICIENT_EVIDENCE** Retrieved APOE4 passages but failed to synthesise the contradiction across them	**UNSUPPORTED ✓** Evidence Reviewer flagged multiple passages directly contradicting the claim *Agent strength: contradiction detection*
"Calpain-2 is involved in cell spreading." Expected: **SUPPORTED**	**SUPPORTED ✓** Direct evidence retrieved, correct verdict in 8.4s — no need for more	**SUPPORTED ✓** Same result via simple path (≤25 words, Smart Gate), took ~20s *RAG sufficiency: simple, direct claims*
"Gastric lavage is effective for paraquat poisoning." Expected: **INSUFFICIENT_EVIDENCE**	**UNSUPPORTED ✗** Over-interpreted tangential toxicity evidence — corpus lacks specific studies	**UNSUPPORTED ✗** Identical error — S2 found no better coverage. Corpus ceiling. *Corpus ceiling: no architecture can fix missing evidence*

E9c: What We Fixed, What We Tested, What It Proves

① WHAT WE FIXED *Three targeted fixes — each addressing a specific root cause from the analysis*

Smart Gate

→ Fixes: Query Dilution

Simple claims (≤25 words) skip decomposition — go direct to ChromaDB. ~20s, 2 LLM calls.

Original-Claim-First

→ Fixes: Query Dilution

Always anchor retrieval on the full original claim. Sub-claims only supplement — not replace.

Aggressive S2 Trigger

→ Fixes: Missed Coverage

External Semantic Scholar search fires on MODERATE evidence — not just WEAK. Catches borderline cases.

② WHAT WE TESTED

E9d: 18 claims on topics NOT in SciFact corpus

These aren't just 'hard' SciFact claims — the topics themselves (e.g. CRISPR off-target effects, mRNA cancer vaccines) were never indexed. **RAG has literally nothing to work with.**

③ WHAT IT PROVES

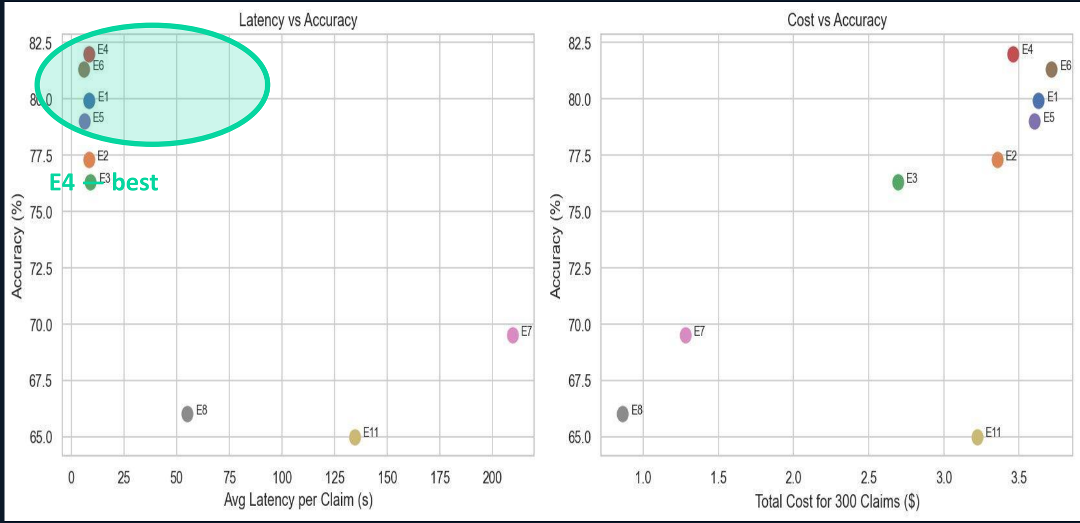
+44.5%

RAG baseline: 22.2% → Agent + S2: 66.7%

Agents justify their cost when the corpus runs out. On in-corpus claims, RAG is still better. Architecture choice should depend on corpus coverage — not be a blanket decision.

Cost & Latency: RAG is Pareto-Optimal on Closed Corpus

On a closed corpus: **spending more money or waiting longer does NOT improve accuracy.** RAG wins on all three axes simultaneously.



E4 (RAG) vs **E7 (Agents)**

8.4s | $3.46 | 82% — ~80s | $5+ | 69.5%

10× slower, 45% more expensive, 13% less accurate

E4 (RAG) vs **E11 (Llama)**

8.4s | $3.46 | 82% — ~12s | $0 local | 65%

Free local model saves cost but loses 17% accuracy

Production Recommendation — Route, Don't Choose:

T1 — RAG first
8.4s · $0.01/claim · 82%

→ *Trigger: RAG confidence low*

T2 — Smart Agent
20–80s · $0.02–0.05/claim

→ *Trigger: local evidence weak/missing*

T3 — External Search
+60s · Semantic Scholar 200M+

Key Findings Summary

01 Chunking > Retrieval Method

Recursive (82%) vs Semantic (76%) = 6pp gap. Naive vs Hybrid = 3pp. Invest in how you split before how you search.

02 Simplicity Wins on Closed Corpora

Best pipeline is also fastest (8.4s) and cheapest. Multi-agent complexity reduced accuracy by 16–17pp on this benchmark.

03 Agents Excel at Contradiction Detection

UNSUPPORTED recall: agents 83-84% ≈ RAG. But SUPPORTED collapses 88% ⟶ 49%. Multi-step review is a built-in devil's advocate.

04 External Search is Context-Dependent

S2 adds noise on in-corpus claims. On out-of-corpus: +44.5pp. Value depends entirely on corpus coverage.

05 Model Quality > Architectural Complexity

Claude Sonnet RAG (82%) vs Llama RAG (65%) = 19pp — bigger lever than architecture choice (17pp).

06 Corpus-Level Ceiling

97.6% of E9c's persistent failures were identical to E4 failures. No architecture fixes missing evidence — motivates tiered approach.

What We Didn't Measure — And Why It Matters

Our metric: **verdict accuracy (SUP / UNS / INS).** But agents also produce step-by-step reasoning and evidence citations. We measured none of that.

The Gap in Our Evaluation

Agents output: decomposed sub-claims, per-sub-claim evidence, contradiction flags, evidence quality ratings (RCT vs observational), and a structured evidence review — then a verdict.

We evaluated only the final label. A correct verdict that cites the wrong study is a different kind of failure — and we have no measure of it.

Why This Could Strengthen the Agent Case

In real health fact-checking — clinical, regulatory, public health — the **explanation is often as important as the verdict.** "UNSUPPORTED because Study X found the opposite effect in a cohort of 10,000" is far more useful than a label alone.

Our accuracy numbers may understate agent value. On verdict accuracy alone, RAG wins. But agents are architecturally built for explainable reasoning — exactly what LLM-as-judge would evaluate.

Study Limitations

- SciFact claims are concise & well-scoped — real misinformation is vague, emotional, multi-faceted
- Out-of-corpus test is small (18 claims) — directionally strong but not conclusive
- Single embedding model — domain-adapted retrieval may significantly change results

What's Actually Next

- **LLM-as-Judge:** Evaluate explanation quality and evidence grounding — the metric that could rebalance RAG vs agents
- **Confidence Gating:** Auto-route between tiers using retrieval confidence — make the 3-tier system self-directing
- **Real-World Claims:** PUBHEALTH / social media — test against the actual problem, not a clean benchmark

?

Thank you